



Deep Learning

Introduction



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https://github.com/safayani/deep_learning_course



Why Do We Need Neural Networks?

- The Limits of Traditional Programming
- **The Vision Challenge:**
 - Writing hand-crafted rules to recognize 3D objects under varying viewpoints, novel lighting, and cluttered backgrounds is practically impossible.
- **The “Black Box” of Human Cognition:**
 - We cannot write down an explicit algorithm for perception because we don’t consciously understand how our own brains process sensory input.
- **Intractable Complexity:**
 - Even if heuristic rules could be formulated, the resulting code would be horrendously complex and fragile.



The Machine Learning Paradigm

- From Explicit Rules to Data-Driven Programs

Traditional Programming	Machine Learning Approach
input: Data + Hand-written Rules → Output: Answers	Input: Data + Desired Outputs → Output: Learned Program / Model

- **Data-Driven Synthesis:**
 - Instead of coding logic by hand, we collect labeled examples ($X \rightarrow Y$) and let the learning algorithm synthesize the mapping function.
- **Non-Traditional Programs:**
 - The learned “program” consists of millions (or billions) of numerical parameters rather than human-readable code lines.
- **Generalization & Adaptability:**
 - **Generalization:** Performs accurately on novel, unseen data.
 - **Retrainability:** Adapts to shifting patterns simply by re-training on new data.
- **Economic Advantage:**
 - Scaling compute is now vastly cheaper than paying engineers to hand-craft task-specific heuristic programs



Standard Benchmark: The MNIST Dataset

- **What is MNIST?**

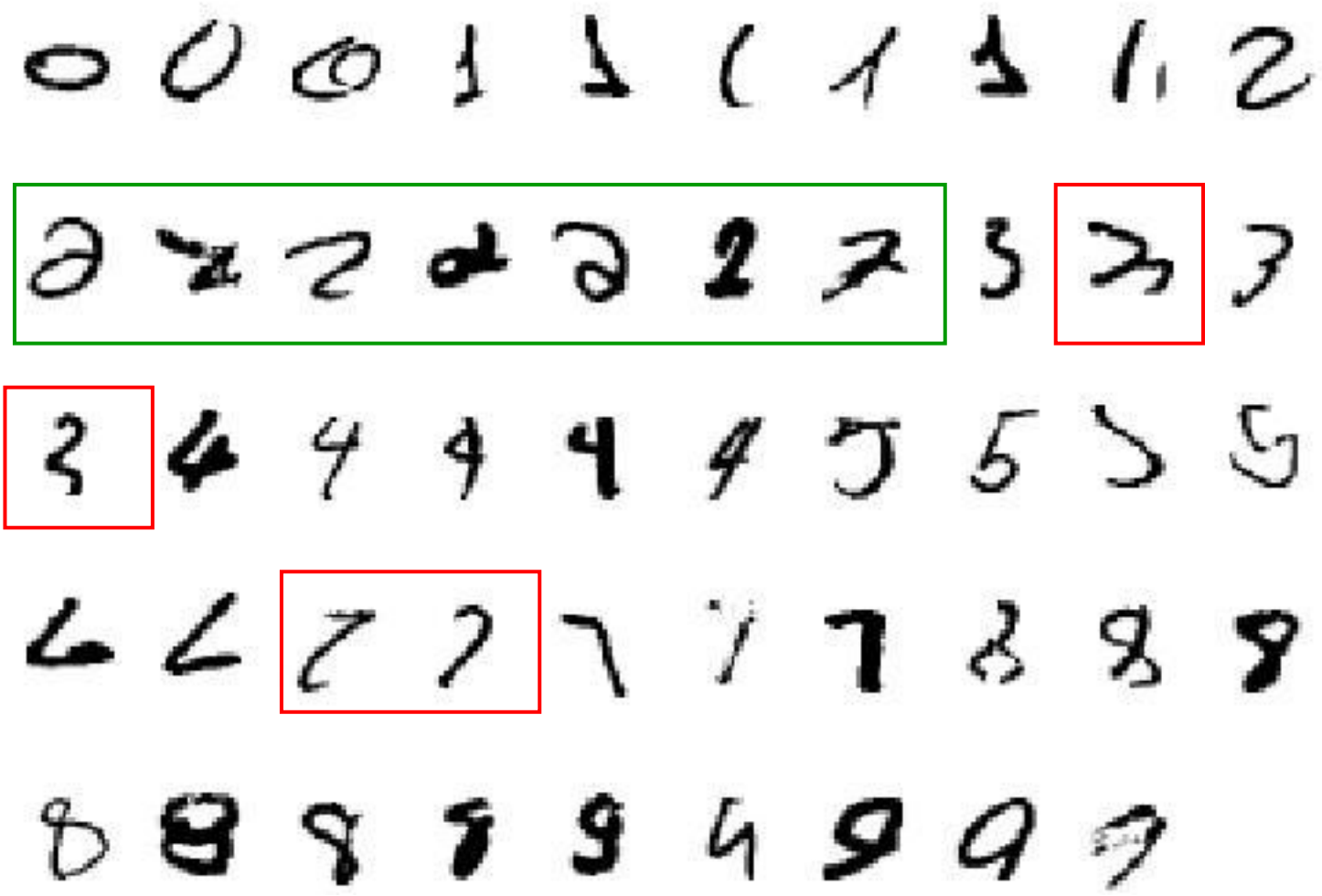
- A canonical dataset of 28×28 pixel grayscale handwritten digits (0–9).

- **Why MNIST?**

- **The Standard Model Organism:** Like fruit flies in genetics, it serves as the universal testing ground for learning algorithms.
 - **Accessible & Fast:** Publicly available and trainable in minutes on moderate hardware / neural networks.
 - **Extensively Benchmarked:** Provides a well-understood performance baseline across decades of machine learning methods.



It is very hard to say what makes a 2

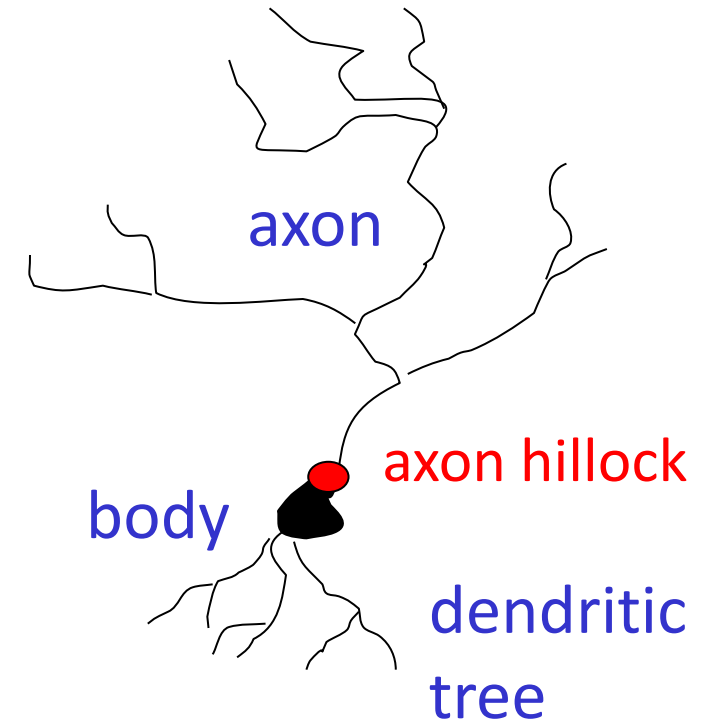


Neural Networks

What are neural networks?

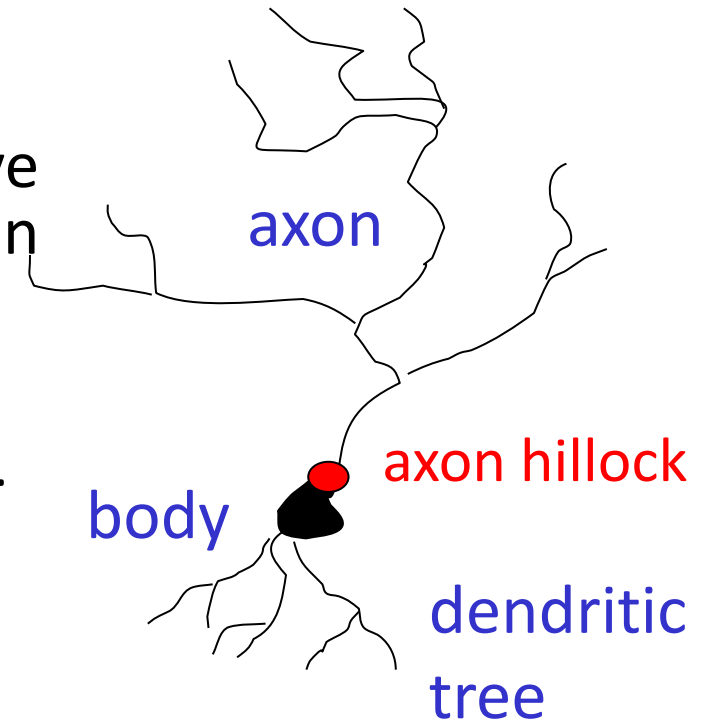
A Typical Cortical Neuron

- **1. Gross Physical Structure**
- **Dendritic Tree:** A branching structure that collects incoming inputs from thousands of other neurons.
- **Axon:** A single output transmission line that branches out to send signals to downstream cells.
- **Synaptic Contacts:** Axons connect to dendritic trees at junctions called **synapses**.
 - A spike arriving at a synapse injects electrical charge (depolarizing or hyperpolarizing) into the receiving cell.



A Typical Cortical Neuron

- **2. Integration & Spike Generation**
- **Charge Accumulation:** Incoming charges from all active synapses flow through the dendrites and accumulate in the cell body.
- **The Axon Hillock (The Comparator):**
 - Located at the junction between the cell body and the axon.
 - It continuously measures the accumulated charge against a critical **firing threshold** (typically around -55 mV).
- **All-or-None Firing:**
 - If accumulated charge $\geq$ **Threshold (θ)**: The neuron fires an outgoing spike (action potential) down the axon.
 - If accumulated charge $<$ **Threshold (θ)**: Nothing happens; the charge leaks away.

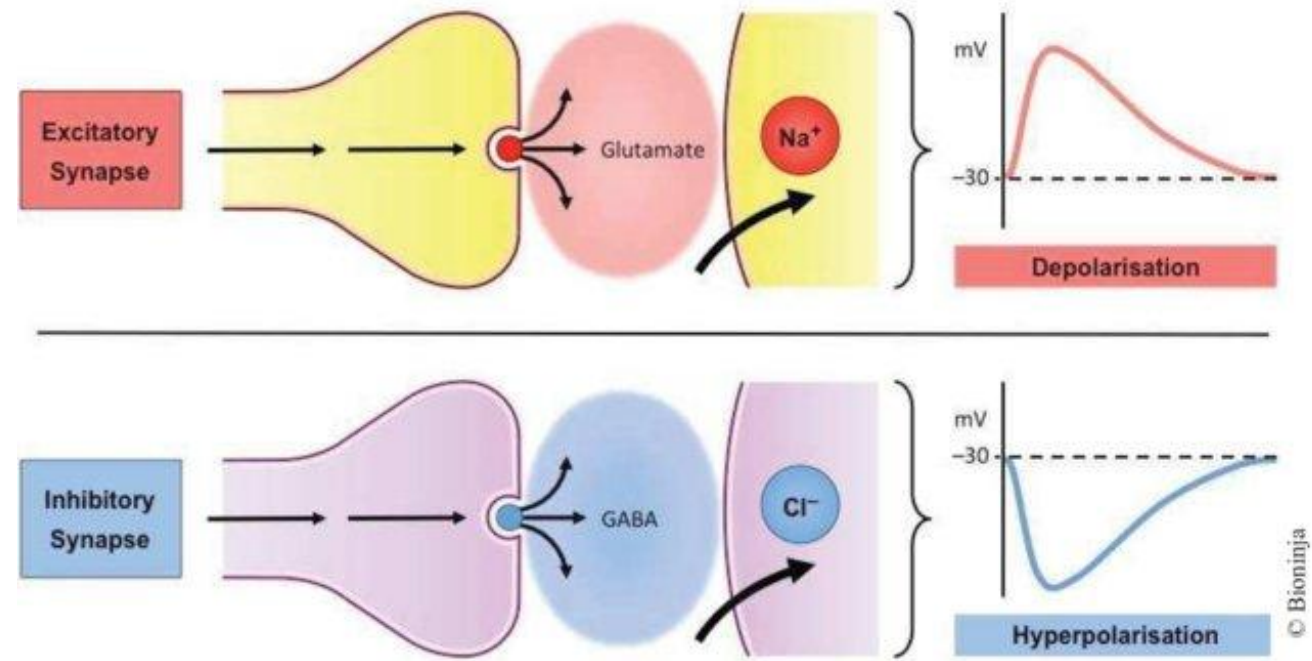
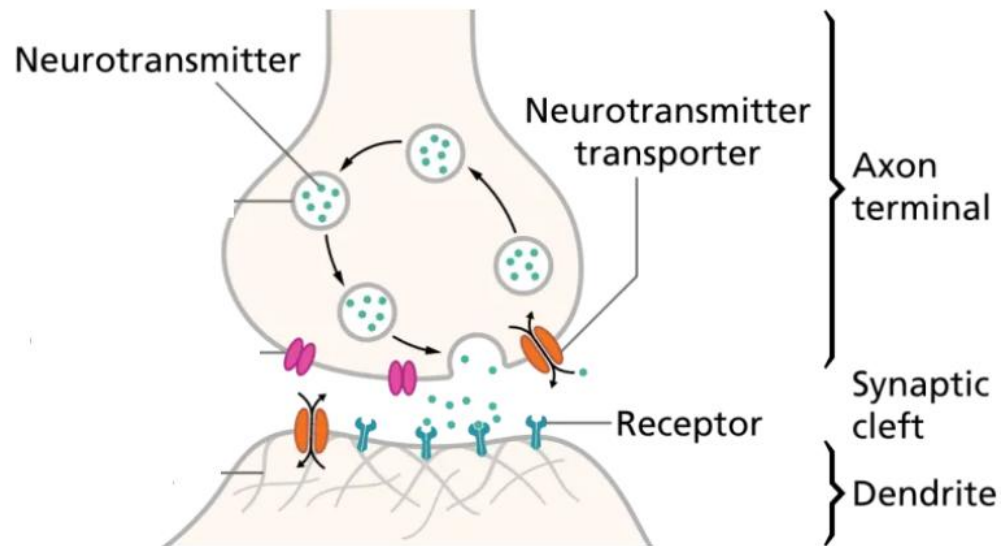


Synaptic Transmission

- **How Information Bridges the Gap Between Neurons**
- **Arrival of the Action Potential**
 - A voltage spike reaches the presynaptic axon terminal.
 - Triggers the release of chemical transmitters stored in synaptic vesicles.
- **Crossing the Synaptic Cleft**
 - Multiple classes of neurotransmitters diffuse across the microscopic junction (≈ 20 nm).
- **Gating Ion Channels**
 - Transmitter molecules bind to target receptors on the postsynaptic membrane.
 - Receptors change shape, opening ion channels that alter the post-synaptic cell's electrical potential.
- **Signal Conversion:** Electrical Signal $\rightarrow$ Chemical Diffusion $\rightarrow$ Electrical Response



Synapses



How synapses adapt

- The effectiveness of the synapse can be changed:
 - vary the number of vesicles of transmitter.
 - vary the number of receptor molecules.
- Synapses are slow, but they have advantages over RAM
 - They are very small and very low-power.
 - They adapt using locally available signals
 - **But what rules do they use to** decide how to change?



How Synapses Adapt (Synaptic Plasticity)

- **1. Modifying Connection Strength (Synaptic Weight)**
- A synapse is not a fixed wire; its transmission strength can be turned up or down:
- **Sending Side (Presynaptic):**
 - Can change the **amount of transmitter** released per spike (more or fewer vesicles).
- **Receiving Side (Postsynaptic):**
 - Can increase or decrease the **number of receptor channels** embedded in its membrane.
- **Result:**
 - **Strong connection:** Releases more chemical → catches more ions → produces a large voltage jump.
 - **Weak connection:** Releases minimal chemical → fewer receptors → produces a negligible voltage jump.



How the Brain Works (On One Slide!)

1. Architecture & Signaling

- **Massive Interconnectivity:** Each neuron receives inputs from thousands of other neurons;
- **Digital Communication:** Cortical neurons communicate via discrete, all-or-nothing electrical pulses called **spikes** (action potentials).

2. Synaptic Weights as Computation

- **Synaptic Scaling:** The effect of each incoming spike is modulated by a **synaptic weight** (w_i).
- **Sign Matters:**
 - **Positive** ($+w$): Excitatory (depolarizes towards firing threshold).
 - **Negative** ($-w$): Inhibitory (hyperpolarizes away from threshold).
- **Plasticity & Learning:** Synaptic weights adapt over time so the network self-organizes to execute complex computations:
 - *Object recognition, speech comprehension, motor control, and planning.*

3. Biological Scale vs. Modern Hardware

- You have about 10^{11} (100 billion) neurons, each with an average of 10^4 (10,000) synaptic connections.
- Total parameter count: $\sim 10^{14}$ to 10^{15} (**hundreds of trillions**) of adaptive weights.
- **Massive Parallel Bandwidth:** Hundreds of trillions of weights operate in parallel within milliseconds, operating at roughly **20 Watts** of power—far outperforming modern workstations in energy efficiency and simultaneous throughput.



Modularity and the brain

- Different bits of the cortex do different things.
 - Local damage to the brain has specific effects.
 - Specific tasks increase the blood flow to specific regions.
- But cortex looks pretty much the same all over.
 - Early brain damage makes functions relocate.
- Cortex is made of general purpose stuff that has the ability to turn into special purpose hardware in response to experience.
 - This gives rapid parallel computation plus flexibility.
 - Conventional computers get flexibility by having stored sequential programs, but this requires very fast central processors to perform long sequential computations.



Object Recognition

PASCAL Visual Object Challenge (20 object categories)

[Everingham et al. 2006-2012]

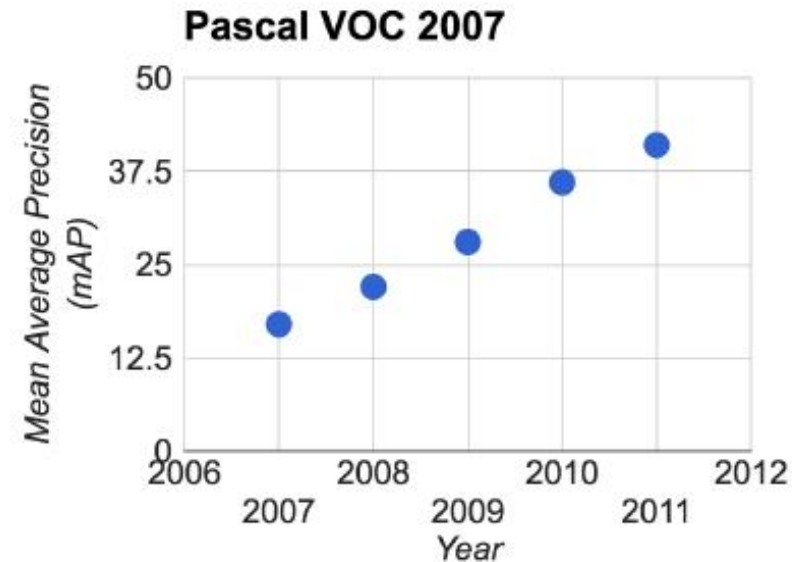
Image is CC0 1.0 public domain



Image is CC0 1.0 public domain



Image is CC0 1.0 public domain





www.image-net.org

22K categories and **14M** images

- Animals
 - Bird
 - Fish
 - Mammal
 - Invertebrate
- Plants
 - Tree
 - Flower
- Food
- Materials
- Structures
 - Artifact
 - Tools
 - Appliances
 - Structures
- Person
- Scenes
 - Indoor
 - Geological Formations
- Sport Activities

IMAGENET Large Scale Visual Recognition Challenge

The Image Classification Challenge:
1,000 object classes
1,431,167 images



Output:
Scale
T-shirt
Steel drum
Drumstick
Mud turtle



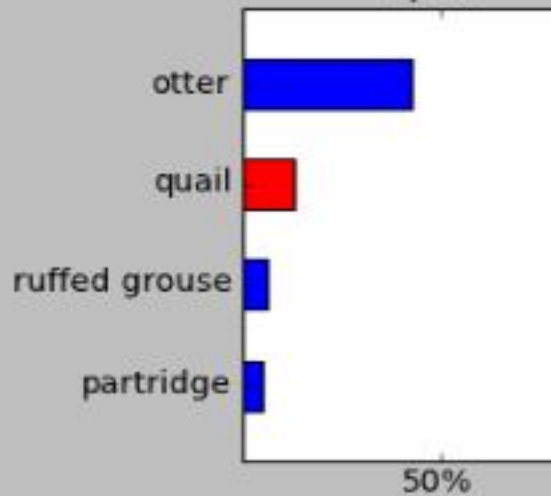
Output:
Scale
T-shirt
Giant panda
Drumstick
Mud turtle



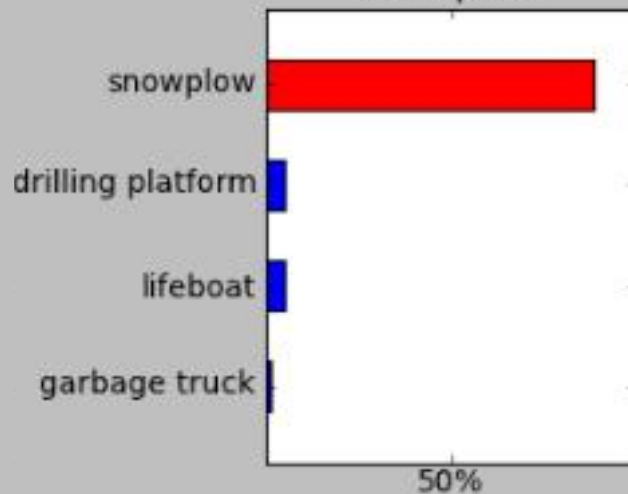
Some examples from an earlier version of the net



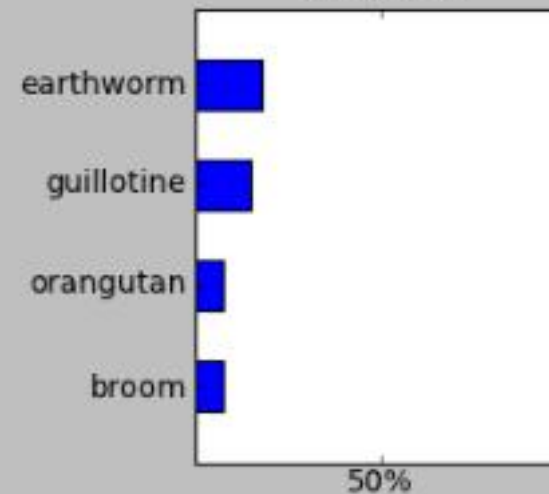
quail



snowplow



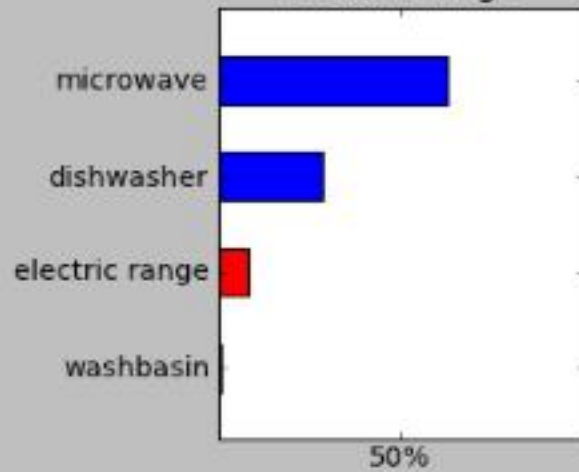
scabbard



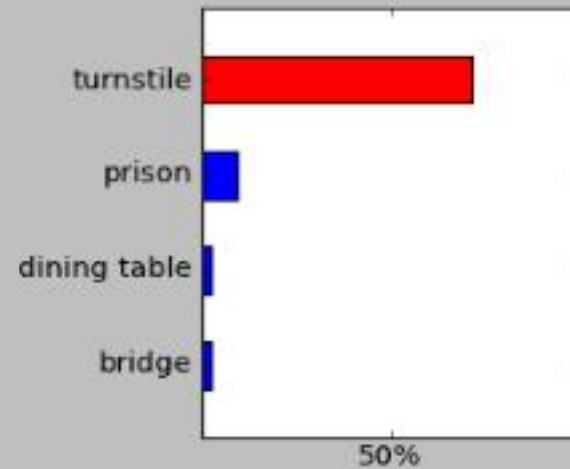
It can deal with a wide range of objects



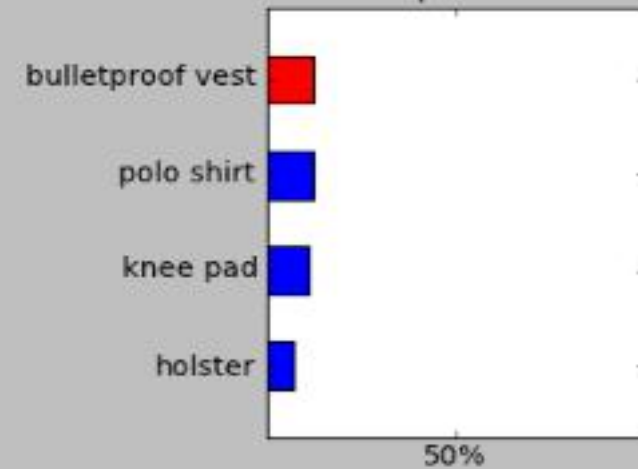
electric range



turnstile

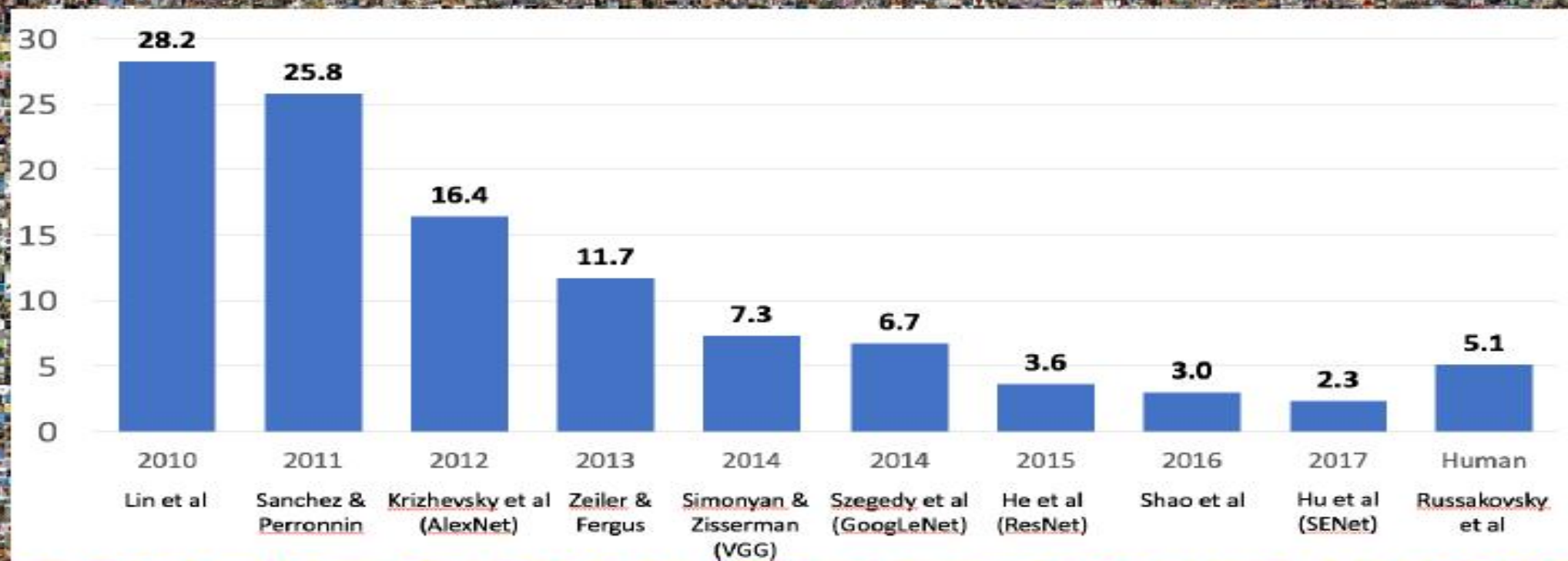


bulletproof vest

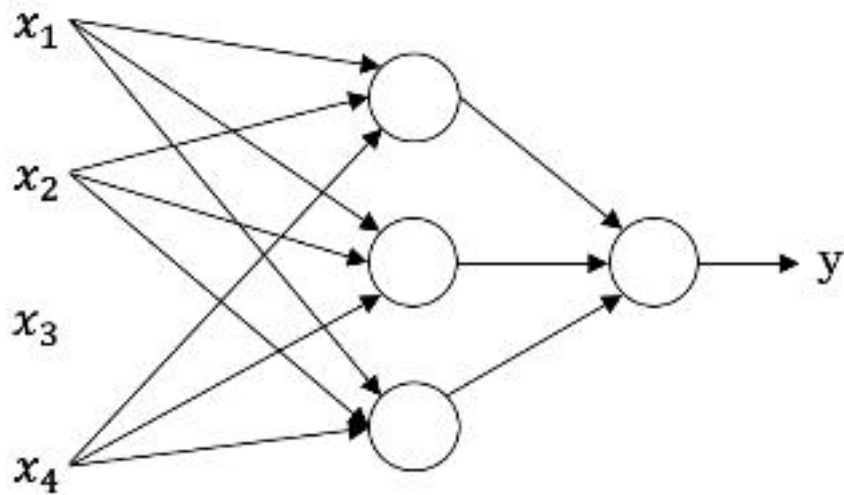


IMAGENET Large Scale Visual Recognition Challenge

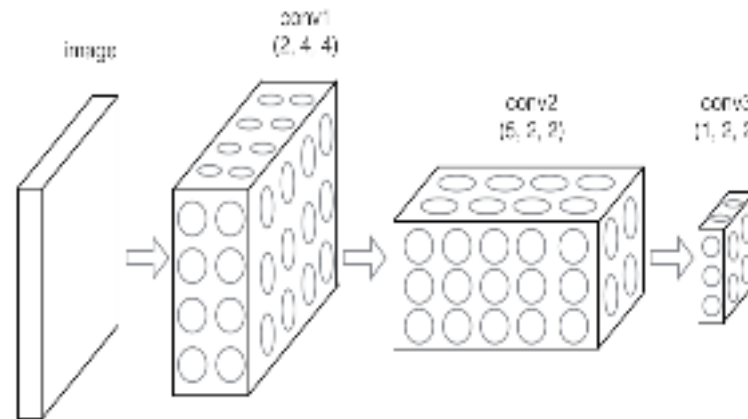
The Image Classification Challenge:
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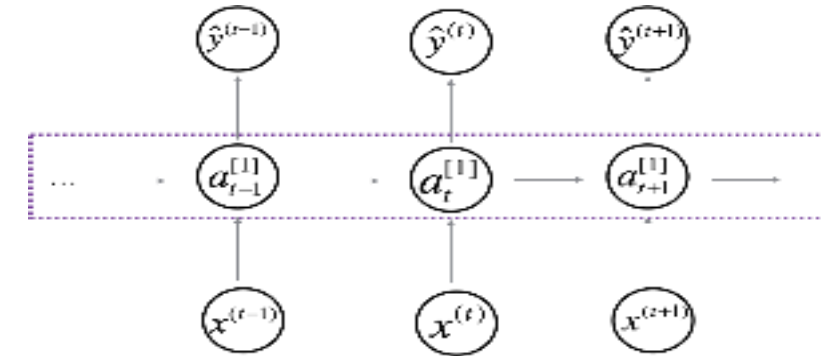
Neural Network examples



Standard NN



Convolutional NN

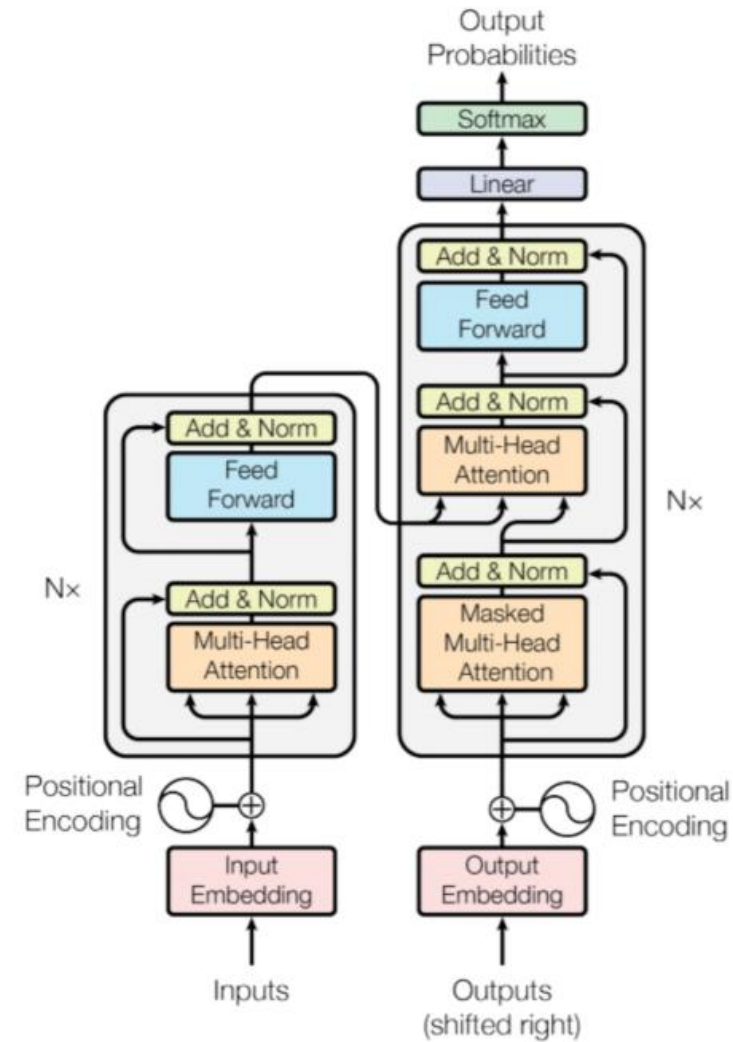


Recurrent NN



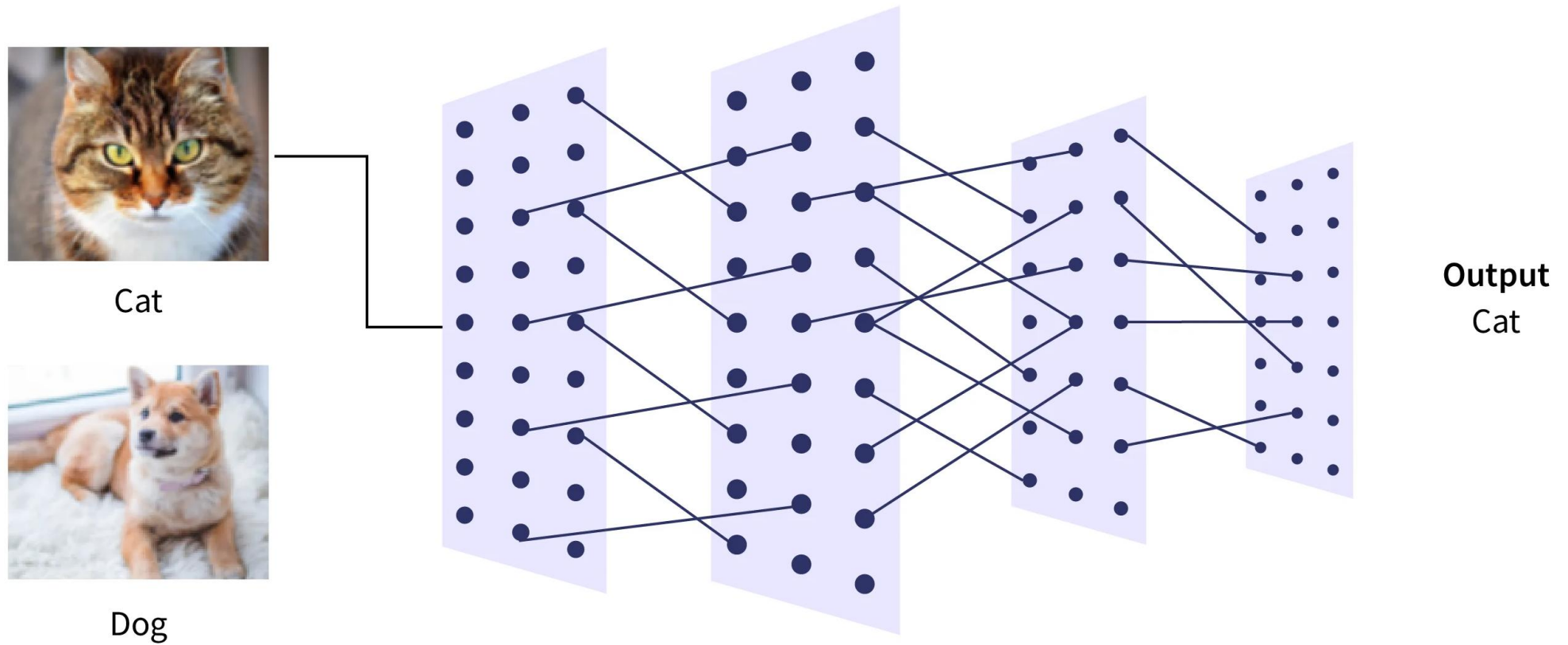
Neural Network examples

Transformers

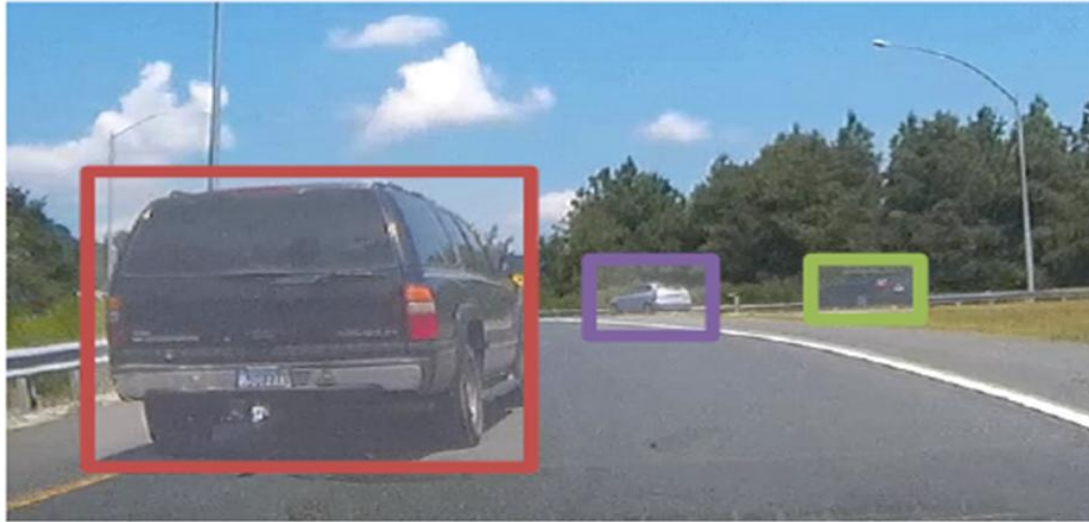


Some applications of Deep Learning

Object Classification



Object Detection



This image is licensed under CC BY-NC-SA 2.0; changes made



Person

Hammer

Neural style transfer



Content



Style



Generated image



Content



Style



Generated image



Image Captioning: Example Results

Captions generated using [neuraltalk2](#)
All images are [CC0 Public domain](#):
[cat suitcase](#), [cat tree](#), [dog](#), [bear](#),
[surfers](#), [tennis](#), [giraffe](#), [motorcycle](#)



A cat sitting on a suitcase on the floor



A cat is sitting on a tree branch



A dog is running in the grass with a frisbee



A white teddy bear sitting in the grass



Two people walking on the beach with surfboards



A tennis player in action on the court



Two giraffes standing in a grassy field



A man riding a dirt bike on a dirt track



Visual Question Answering



Q: What endangered animal is featured on the truck?

A: A bald eagle.

A: A sparrow.

A: A humming bird.

A: A raven.



Q: Where will the driver go if turning right?

A: Onto 24 1/4 Rd.

A: Onto 25 3/4 Rd.

A: Onto 23 3/4 Rd.

A: Onto Main Street.



Q: Who is under the umbrella?

A: Two women.

A: A child.

A: An old man.

A: A husband and a wife.

Natural Language Processing

- Word Embedding
- Sentiment Classification
- Machine Translation
- Question answering, chat-gpt



Sentiment classification problem

x

The dessert is excellent.

Service was quite slow.

Good for a quick meal, but nothing special.

Completely lacking in good taste, good service, and good ambience.

y



Generative Models

- Generative Adversarial Network
- Variational Autoencoder



Image-to-Image Translation

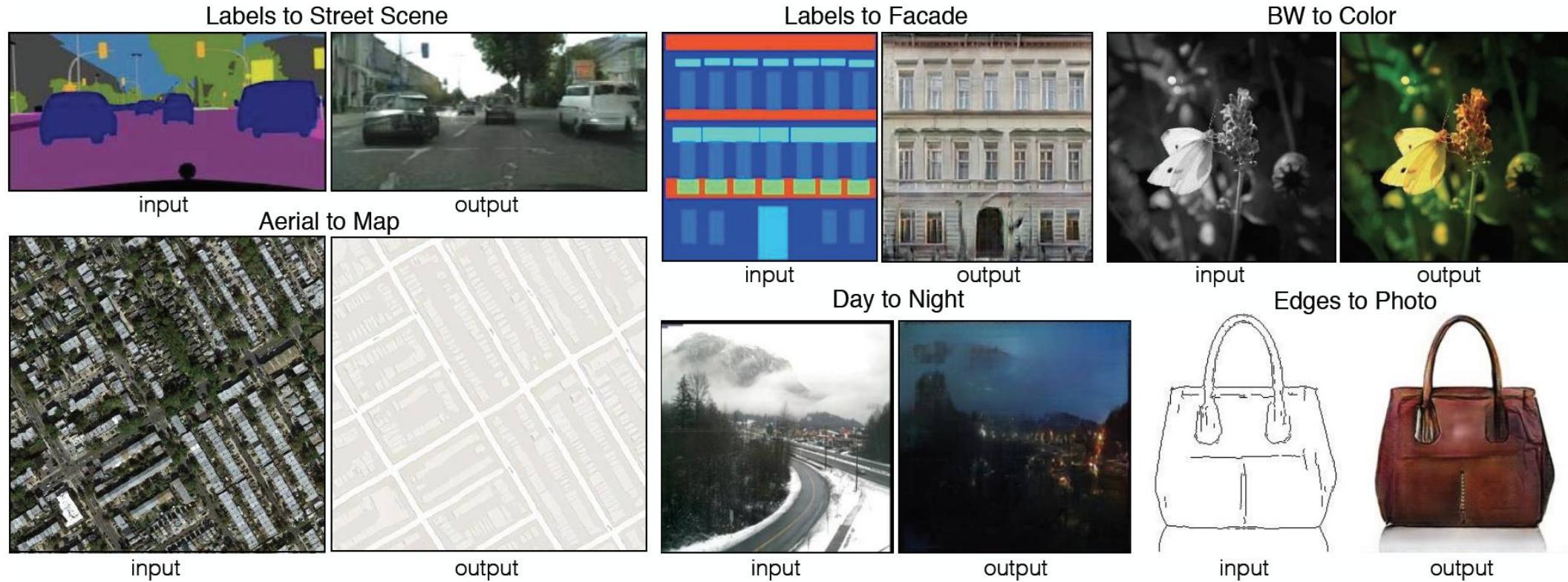


Figure 1 in the original paper.

[Link to an interactive demo of this paper](#)



Text-to-Image Synthesis

Motivation

Given a text description, generate images closely associated.

Uses a conditional GAN with the generator and discriminator being condition on “dense” text embedding.

this small bird has a pink breast and crown, and black primaries and secondaries.



this magnificent fellow is almost all black with a red crest, and white cheek patch.



the flower has petals that are bright pinkish purple with white stigma



this white and yellow flower have thin white petals and a round yellow stamen



Figure 1 in the original paper.

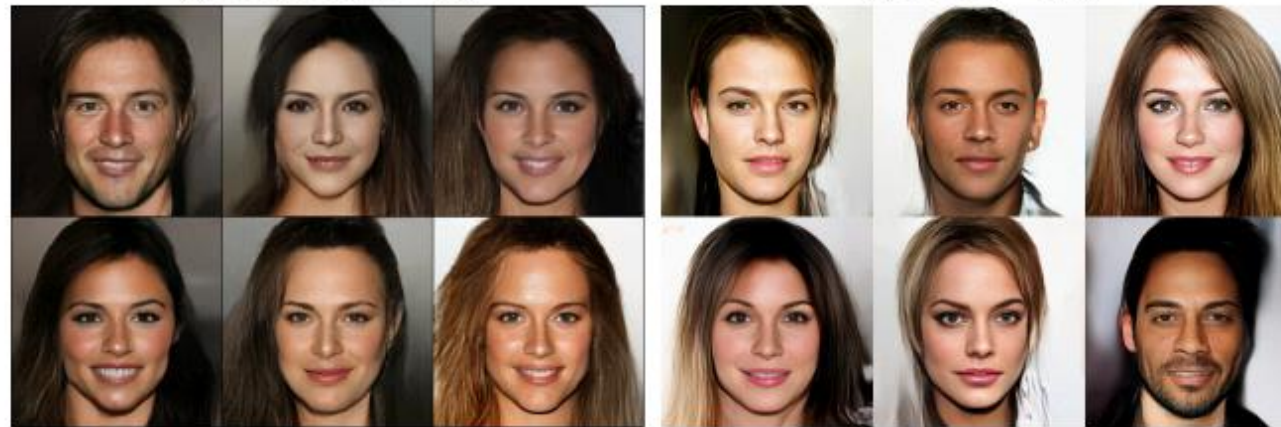


Face images generated with a Variational Autoencoder



(d) CelebA HQ ($t = 0.6$)

(e) FFHQ ($t = 0.5$)



Course outline

- Introduction
- Logistic regression
- Multi-layer perceptron (MLP)
- Back Propagation (BP)
- Regularization concept
- Optimization
- Convolution NNs
- Introducing some famous Convolution NN architectures (LeNet-5, AlexNet, VGG, google net, Resnet, mobile net, efficient net)
- Some applications of deep learning(localization, detection, face recognition, style transfer)
- Recurrent neural network (GRU, LSTM)
- Transformers
- Basic concepts of NLP
- Generative AI (GAN, VAE)



Related Courses

- Deep Learning, Andrew NG.



- Convolutional Neural Networks for Visual Recognition, Fei-Fei Lee, Justin Johnson, Serena Yeung



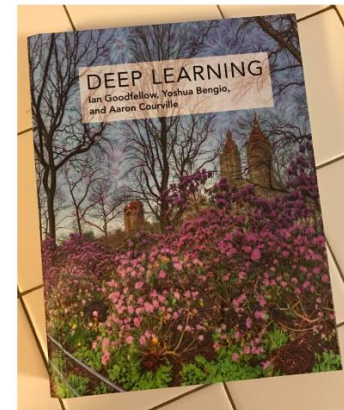
- Neural Network for Machine Learning



Textbooks

- Deep learning, I. Goodfellow, J. Bengio

MIT Press, 2016



- Neural Networks and deep learning, M. Nielsen

Determination Press, 2015



Course Strategy

- Assignments and project 5-7 marks (**python**)
- Midterm and quizzes 5-7 marks
- Final 5-7 marks





۸۱.۲ هزار
بازدید کل

۸۹۳
دنبال کننده

تنظیمات

چیدمان صفحه

هوش مصنوعی برای همه

خانه ویدیوها ویدیوهای کوتاه لیست های پخش درباره کانال

:

یادگیر عمیق ج ۲۷ خودکدگذار تغییراتی (VAE) بخش ۲: مبانی ریاضی و آموزش ۱۴۰۵

چگونه مدل VAE را آموزش دهیم روابط آموزش VAE محاسبه ELBO Evidence Lower Bound تابع هدف حقه بازپارامترسازی Reparameterization trick مثال هایی از تولید داده

۱۲۷ بازدید . ۴ ماه پیش



Deep Learning

Variational Autoencoder (VAE)

Part 2: Mathematical Foundations and Training

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۳۹:۴۰ https://github.com/safayani/deep_learning_course

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